Experiment 7

Aim:To study the Truffle framework, establish a local blockchain, and deploy a smart contract using Truffle.

Software Requirements

- Node.js and npm
- Truffle Suite
- Ganache (local blockchain)
- Visual Studio Code or Terminal
- Solidity programming language

Theory

Truffle is a development framework used to compile, test, and deploy Ethereum smart contracts written in Solidity.

Ganache provides a local blockchain for development and testing. It supplies test accounts with simulated cryptocurrency, allowing smart contracts to be deployed without using real ETH.

A smart contract is a program stored on a blockchain that executes according to predefined rules. Truffle uses a configuration file and migration scripts to deploy contracts to a selected blockchain network.

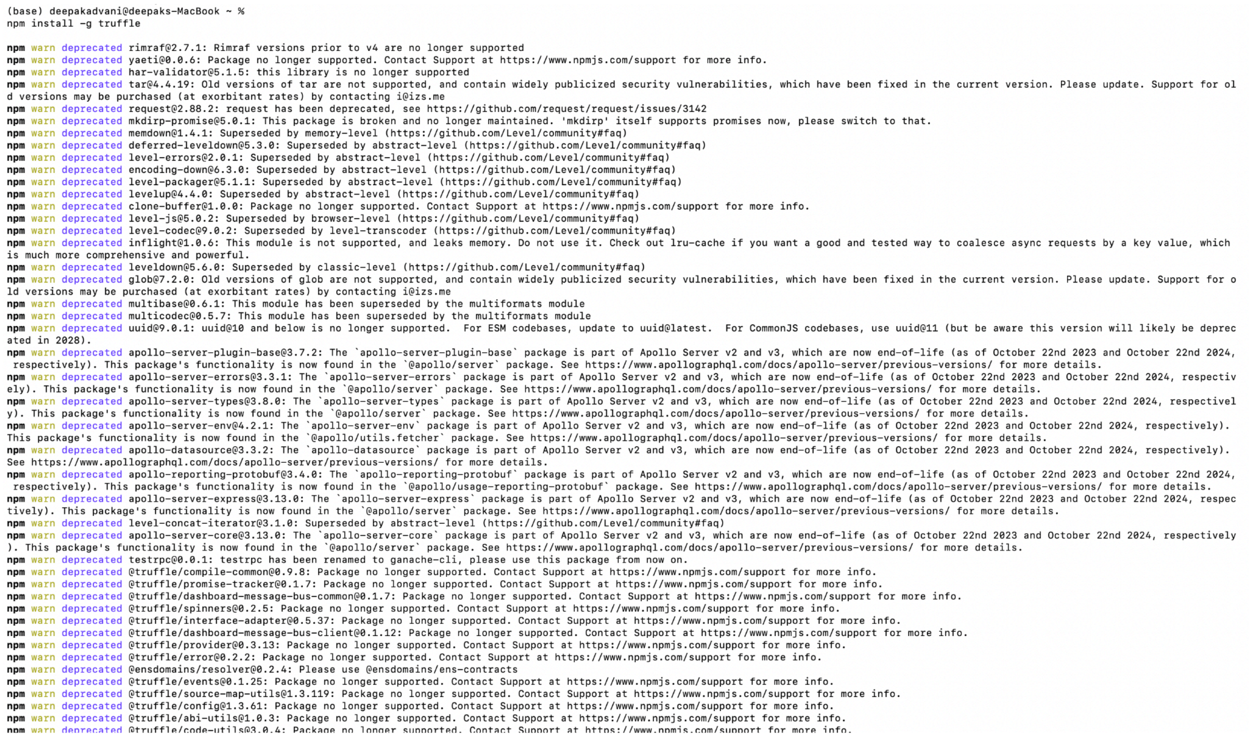
The basic workflow is:

1. Create a Truffle project.
2. Write a Solidity smart contract.
3. Compile the contract.
4. Start a local blockchain.
5. Configure Truffle and deploy the contract.
6. Verify the deployment output.

Step 1: Open Terminal and check softwares required

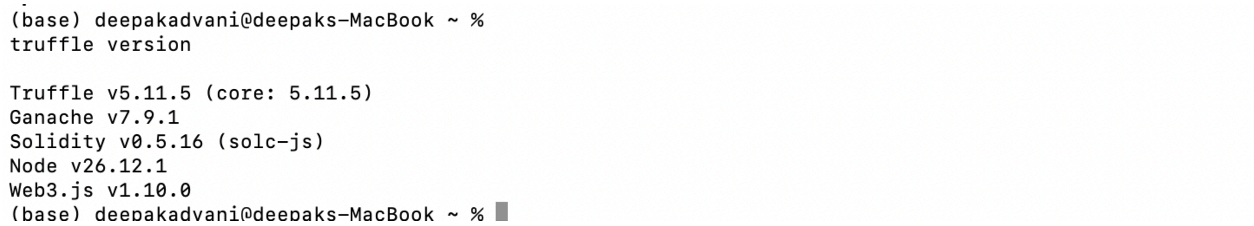
```
(base) deepakadvani@deepaks-MacBook ~ %
node --version

v26.12.1
(base) deepakadvani@deepaks-MacBook ~ %
npm --version

11.20.0
```

Step-2 install truffle



```
(base) deepakadvani@deepaks-MacBook ~ %
npm install -g truffle

npm warn deprecated rimraf@2.7.1: Rimraf versions prior to v4 are no longer supported
npm warn deprecated yaeti@0.0.6: Package no longer supported. Contact Support at https://www.npmjs.com/support for more info.
npm warn deprecated har-validator@5.1.5: this library is no longer supported
npm warn deprecated tar@4.4.19: Old versions of tar are not supported, and contain widely publicized security vulnerabilities, which have been fixed in the current version. Please update. Support for ol
d versions may be purchased (at exorbitant rates) by contacting i@izs.me
npm warn deprecated request@2.88.2: request has been deprecated, see https://github.com/request/request/issues/3142
npm warn deprecated mkdirp-promise@5.0.1: This package is broken and no longer maintained. 'mkdirp' itself supports promises now, please switch to that.
npm warn deprecated memdown@1.4.1: Superseded by memory-level (https://github.com/Level/community#faq)
npm warn deprecated deferred-leveldown@5.3.0: Superseded by abstract-level (https://github.com/Level/community#faq)
npm warn deprecated level-errors@2.0.1: Superseded by abstract-level (https://github.com/Level/community#faq)
npm warn deprecated encoding-down@6.3.0: Superseded by abstract-level (https://github.com/Level/community#faq)
npm warn deprecated level-packager@5.1.1: Superseded by abstract-level (https://github.com/Level/community#faq)
npm warn deprecated levelup@4.4.0: Superseded by abstract-level (https://github.com/Level/community#faq)
npm warn deprecated clone-buffer@1.0.0: Package no longer supported. Contact Support at https://www.npmjs.com/support for more info.
npm warn deprecated level-js@5.0.2: Superseded by browser-level (https://github.com/Level/community#faq)
npm warn deprecated level-codec@9.0.2: Superseded by level-transcoder (https://github.com/Level/community#faq)
npm warn deprecated inflight@1.0.6: This module is not supported, and leaks memory. Do not use it. Check out lru-cache if you want a good and tested way to coalesce async requests by a key value, which
is much more comprehensive and powerful.
npm warn deprecated leveldown@5.6.0: Superseded by classic-level (https://github.com/Level/community#faq)
npm warn deprecated glob@7.2.0: Old versions of glob are not supported, and contain widely publicized security vulnerabilities, which have been fixed in the current version. Please update. Support for o
ld versions may be purchased (at exorbitant rates) by contacting i@izs.me
npm warn deprecated multibase@0.6.1: This module has been superseded by the multiformats module
npm warn deprecated multicodec@0.5.7: This module has been superseded by the multiformats module
npm warn deprecated uuid@9.0.1: uuid@10 and below is no longer supported.  For ESM codebases, update to uuid@latest.  For CommonJS codebases, use uuid@11 (but be aware this version will likely be deprec
ated in 2028).
npm warn deprecated apollo-server-plugin-base@3.7.2: The `apollo-server-plugin-base` package is part of Apollo Server v2 and v3, which are now end-of-life (as of October 22nd 2023 and October 22nd 2024,
 respectively). This package's functionality is now found in the `@apollo/server` package. See https://www.apollographql.com/docs/apollo-server/previous-versions/ for more details.
npm warn deprecated apollo-server-errors@3.3.1: The `apollo-server-errors` package is part of Apollo Server v2 and v3, which are now end-of-life (as of October 22nd 2023 and October 22nd 2024, respectiv
ely). This package's functionality is now found in the `@apollo/server` package. See https://www.apollographql.com/docs/apollo-server/previous-versions/ for more details.
npm warn deprecated apollo-server-types@3.8.0: The `apollo-server-types` package is part of Apollo Server v2 and v3, which are now end-of-life (as of October 22nd 2023 and October 22nd 2024, respectivel
y). This package's functionality is now found in the `@apollo/server` package. See https://www.apollographql.com/docs/apollo-server/previous-versions/ for more details.
npm warn deprecated apollo-server-env@4.2.1: The `apollo-server-env` package is part of Apollo Server v2 and v3, which are now end-of-life (as of October 22nd 2023 and October 22nd 2024, respectively).
This package's functionality is now found in the `@apollo/utils.fetcher` package. See https://www.apollographql.com/docs/apollo-server/previous-versions/ for more details.
npm warn deprecated apollo-datasource@3.3.2: The `apollo-datasource` package is part of Apollo Server v2 and v3, which are now end-of-life (as of October 22nd 2023 and October 22nd 2024, respectively).
See https://www.apollographql.com/docs/apollo-server/previous-versions/ for more details.
npm warn deprecated apollo-reporting-protobuf@3.4.0: The `apollo-reporting-protobuf` package is part of Apollo Server v2 and v3, which are now end-of-life (as of October 22nd 2023 and October 22nd 2024,
 respectively). This package's functionality is now found in the `@apollo/usage-reporting-protobuf` package. See https://www.apollographql.com/docs/apollo-server/previous-versions/ for more details.
npm warn deprecated apollo-server-express@3.13.0: The `apollo-server-express` package is part of Apollo Server v2 and v3, which are now end-of-life (as of October 22nd 2023 and October 22nd 2024, respec
tively). This package's functionality is now found in the `@apollo/server` package. See https://www.apollographql.com/docs/apollo-server/previous-versions/ for more details.
npm warn deprecated level-concat-iterator@3.1.0: Superseded by abstract-level (https://github.com/Level/community#faq)
npm warn deprecated apollo-server-core@3.13.0: The `apollo-server-core` package is part of Apollo Server v2 and v3, which are now end-of-life (as of October 22nd 2023 and October 22nd 2024, respectively
). This package's functionality is now found in the `@apollo/server` package. See https://www.apollographql.com/docs/apollo-server/previous-versions/ for more details.
npm warn deprecated testrpc@0.0.1: testrpc has been renamed to ganache-cli, please use this package from now on.
npm warn deprecated @truffle/compile-common@0.9.8: Package no longer supported. Contact Support at https://www.npmjs.com/support for more info.
npm warn deprecated @truffle/promise-tracker@0.1.7: Package no longer supported. Contact Support at https://www.npmjs.com/support for more info.
npm warn deprecated @truffle/dashboard-message-bus-common@0.1.7: Package no longer supported. Contact Support at https://www.npmjs.com/support for more info.
npm warn deprecated @truffle/spinners@0.2.5: Package no longer supported. Contact Support at https://www.npmjs.com/support for more info.
npm warn deprecated @truffle/interface-adapter@0.5.37: Package no longer supported. Contact Support at https://www.npmjs.com/support for more info.
npm warn deprecated @truffle/dashboard-message-bus-client@0.1.12: Package no longer supported. Contact Support at https://www.npmjs.com/support for more info.
npm warn deprecated @truffle/provider@0.3.13: Package no longer supported. Contact Support at https://www.npmjs.com/support for more info.
npm warn deprecated @truffle/error@0.2.2: Package no longer supported. Contact Support at https://www.npmjs.com/support for more info.
npm warn deprecated @ensdomains/resolver@0.2.4: Please use @ensdomains/ens-contracts
npm warn deprecated @truffle/events@0.1.25: Package no longer supported. Contact Support at https://www.npmjs.com/support for more info.
npm warn deprecated @truffle/source-map-utils@1.3.119: Package no longer supported. Contact Support at https://www.npmjs.com/support for more info.
npm warn deprecated @truffle/config@1.3.61: Package no longer supported. Contact Support at https://www.npmjs.com/support for more info.
npm warn deprecated @truffle/abi-utils@1.0.3: Package no longer supported. Contact Support at https://www.npmjs.com/support for more info.
npm warn deprecated @truffle/code-utils@3.0.4: Package no longer supported. Contact Support at https://www.npmjs.com/support for more info.
```

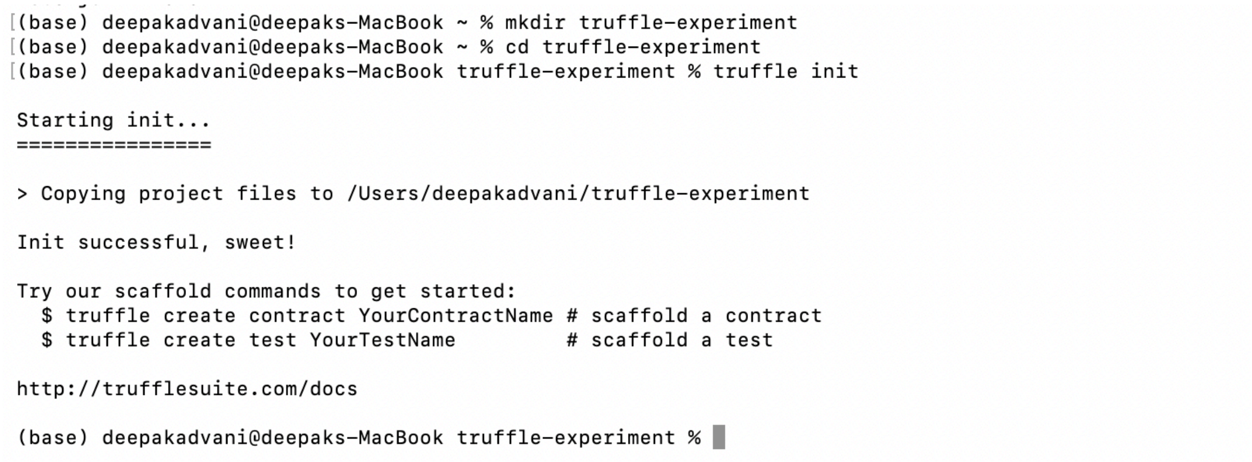
Step-3 check version of truffle



```
(base) deepakadvani@deepaks-MacBook ~ %
truffle version

Truffle v5.11.5 (core: 5.11.5)
Ganache v7.9.1
Solidity v0.5.16 (solc-js)
Node v26.12.1
Web3.js v1.10.0
(base) deepakadvani@deepaks-MacBook ~ %
```

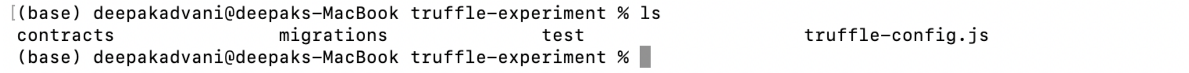
Step 4: Create a Truffle project

1. Create a project folder
2. Enter the folder
3. Initialize the Truffle project



```
[(base) deepakadvani@deepaks-MacBook ~ % mkdir truffle-experiment
[(base) deepakadvani@deepaks-MacBook ~ % cd truffle-experiment
[(base) deepakadvani@deepaks-MacBook truffle-experiment % truffle init

Starting init...
================

> Copying project files to /Users/deepakadvani/truffle-experiment

Init successful, sweet!

Try our scaffold commands to get started:
  $ truffle create contract YourContractName # scaffold a contract
  $ truffle create test YourTestName         # scaffold a test

http://trufflesuite.com/docs

(base) deepakadvani@deepaks-MacBook truffle-experiment % 
```

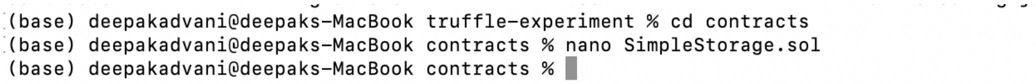
Step 5: Check the project files



```
[(base) deepakadvani@deepaks-MacBook truffle-experiment % ls
contracts               migrations              test                    truffle-config.js
(base) deepakadvani@deepaks-MacBook truffle-experiment % 
```

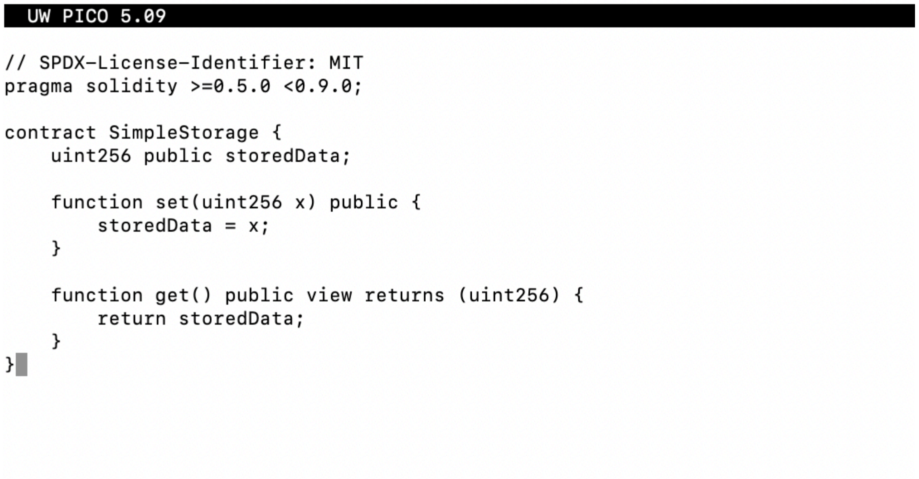
Step 6: Create a Smart Contract

1. Open the contracts folder
2. Create a new contract file



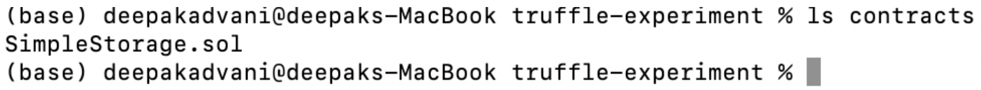
```
(base) deepakadvani@deepaks-MacBook truffle-experiment % cd contracts
(base) deepakadvani@deepaks-MacBook contracts % nano SimpleStorage.sol
(base) deepakadvani@deepaks-MacBook contracts % 
```

Solidity code:



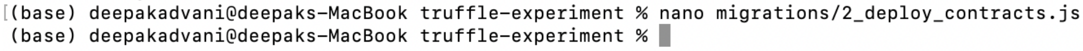
```
  UW PICO 5.09

// SPDX-License-Identifier: MIT
pragma solidity >=0.5.0 <0.9.0;

contract SimpleStorage {
    uint256 public storedData;

    function set(uint256 x) public {
        storedData = x;
    }

    function get() public view returns (uint256) {
        return storedData;
    }
}
```

Step 7: Check the contract file



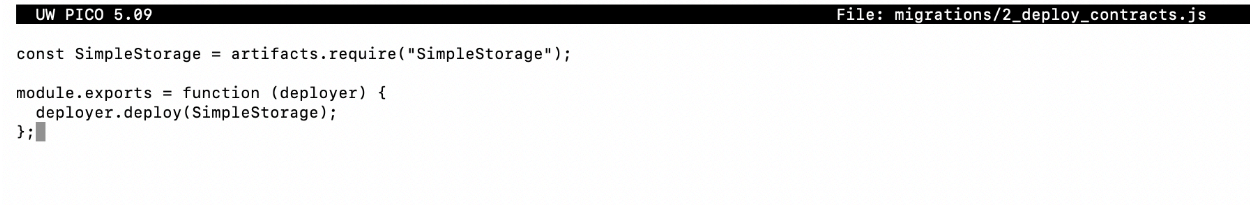
```
(base) deepakadvani@deepaks-MacBook truffle-experiment % ls contracts
SimpleStorage.sol
(base) deepakadvani@deepaks-MacBook truffle-experiment %
```

Step 8: Create the migration file



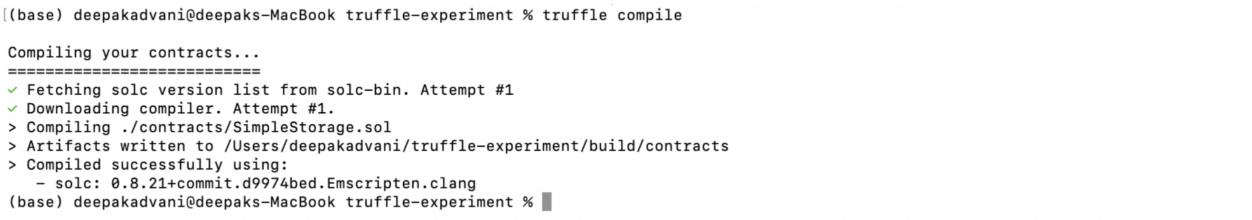
```
[(base) deepakadvani@deepaks-MacBook truffle-experiment % nano migrations/2_deploy_contracts.js
 (base) deepakadvani@deepaks-MacBook truffle-experiment %
```

migration file:



```
const SimpleStorage = artifacts.require("SimpleStorage");

module.exports = function (deployer) {
  deployer.deploy(SimpleStorage);
};
```

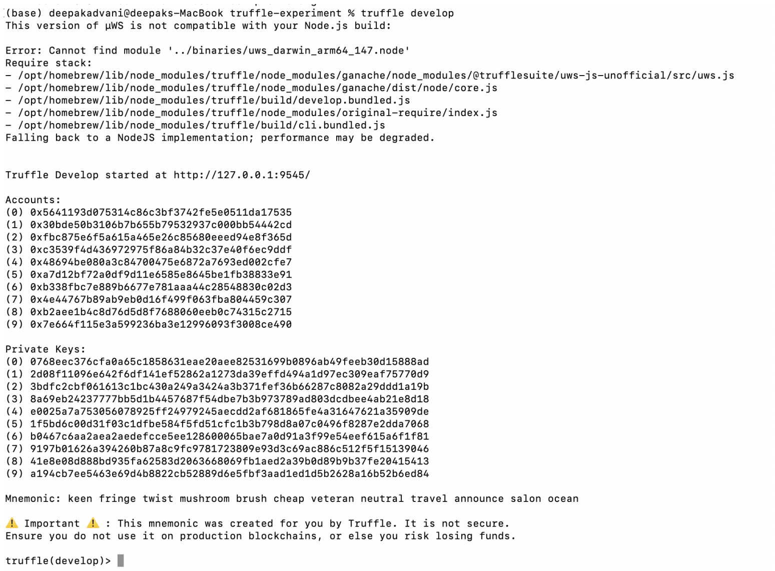
Step 9: Compile the smart contract



```
[(base) deepakadvani@deepaks-MacBook truffle-experiment % truffle compile

Compiling your contracts...
===========================
✓ Fetching solc version list from solc-bin. Attempt #1
✓ Downloading compiler. Attempt #1.
> Compiling ./contracts/SimpleStorage.sol
> Artifacts written to /Users/deepakadvani/truffle-experiment/build/contracts
> Compiled successfully using:
   - solc: 0.8.21+commit.d9974bed.Emscripten.clang
(base) deepakadvani@deepaks-MacBook truffle-experiment %
```

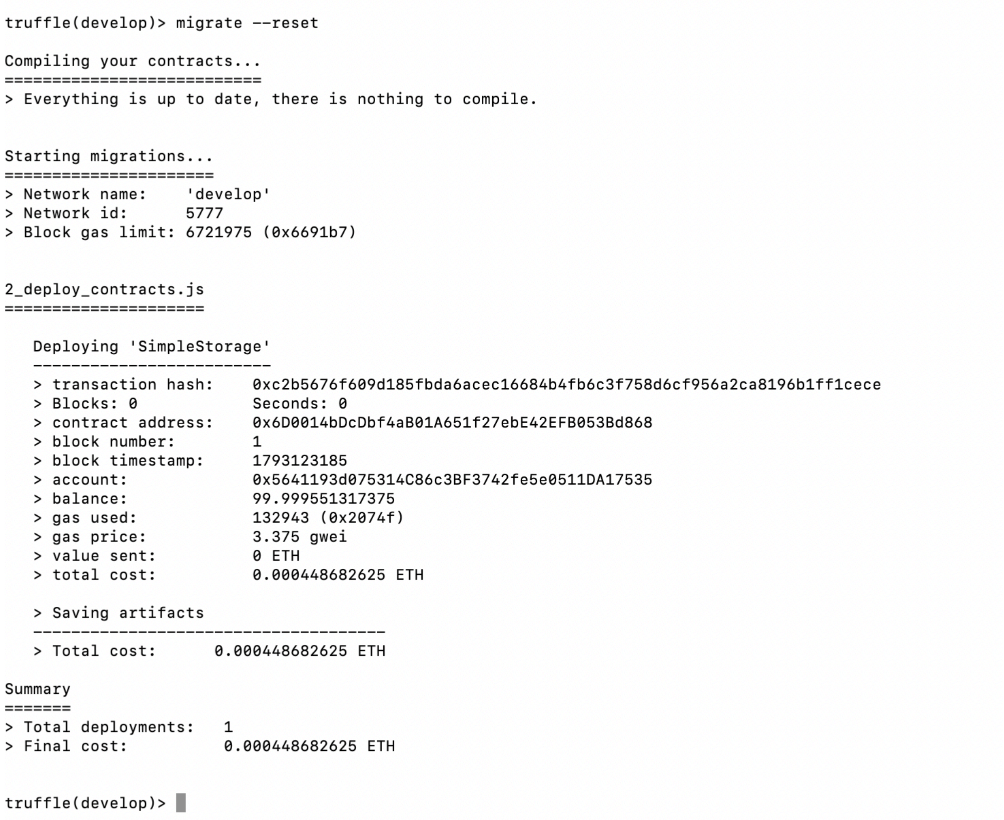
Step 10: Establish a local blockchain

We'll use Truffle's built-in development blockchain, so i don't need to install Ganache separately for this step.



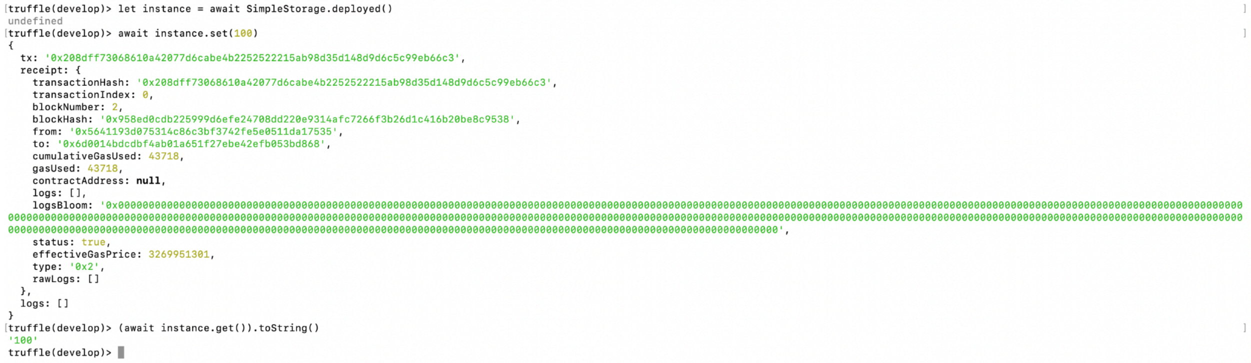
```
(base) deepakadvani@deepaks-MacBook truffle-experiment % truffle develop
This version of µWS is not compatible with your Node.js build:

Error: Cannot find module '../binaries/uws_darwin_arm64_147.node'
Require stack:
- /opt/homebrew/lib/node_modules/truffle/node_modules/ganache/node_modules/@trufflesuite/uws-js-unofficial/src/uws.js
- /opt/homebrew/lib/node_modules/truffle/node_modules/ganache/dist/node/core.js
- /opt/homebrew/lib/node_modules/truffle/build/develop.bundled.js
- /opt/homebrew/lib/node_modules/truffle/node_modules/original-require/index.js
- /opt/homebrew/lib/node_modules/truffle/build/cli.bundled.js
Falling back to a NodeJS implementation; performance may be degraded.


Truffle Develop started at http://127.0.0.1:9545/

Accounts:
(0) 0x5641193d075314c86c3bf3742fe5e0511da17535
(1) 0x30bde50b3106b7b655b79532937c000bb54442cd
(2) 0xfbc875e6f5a615a465e26c85680eeed94e8f365d
(3) 0xc3539f4d436972975f86a84b32c37e40f6ec9ddf
(4) 0x48694be080a3c84700475e6872a7693ed002cfe7
(5) 0xa7d12bf72a0df9d11e6585e8645be1fb38833e91
(6) 0xb338fbc7e889b6677e781aaa44c28548830c02d3
(7) 0x4e44767b89ab9eb0d16f499f063fba804459c307
(8) 0xb2aee1b4c8d76d5d8f7688060eeb0c74315c2715
(9) 0x7e664f115e3a599236ba3e12996093f3008ce490

Private Keys:
(0) 0768eec376cfa0a65c1858631eae20aee82531699b0896ab49feeb30d15888ad
(1) 2d08f11096e642f6df141ef52862a1273da39effd494a1d97ec309eaf75770d9
(2) 3bdfc2cbf061613c1bc430a249a3424a3b371fef36b66287c8082a29ddd1a19b
(3) 8a69eb24237777bb5d1b4457687f54dbe7b3b973789ad803dcdbee4ab21e8d18
(4) e0025a7a753056078925ff24979245aecdd2af681865fe4a31647621a35909de
(5) 1f5bd6c00d31f03c1dfbe584f5fd51cfc1b3b798d8a07c0496f8287e2dda7068
(6) b0467c6aa2aea2aedefcce5ee128600065bae7a0d91a3f99e54eef615a6f1f81
(7) 9197b01626a394260b87a8c9fc9781723809e93d3c69ac886c512f5f15139046
(8) 41e8e08d888bd935fa62583d2063668069fb1aed2a39b0d89b9b37fe20415413
(9) a194cb7ee5463e69d4b8822cb52889d6e5fbf3aad1ed1d5b2628a16b52b6ed84

Mnemonic: keen fringe twist mushroom brush cheap veteran neutral travel announce salon ocean

⚠️ Important ⚠️ : This mnemonic was created for you by Truffle. It is not secure.
Ensure you do not use it on production blockchains, or else you risk losing funds.

truffle(develop)>
```

Step 11: Deploy the smart contract



```
truffle(develop)> migrate --reset

Compiling your contracts...
===========================
> Everything is up to date, there is nothing to compile.


Starting migrations...
======================
> Network name:    'develop'
> Network id:      5777
> Block gas limit: 6721975 (0x6691b7)


2_deploy_contracts.js
=====================

   Deploying 'SimpleStorage'
   -------------------------
   > transaction hash:    0xc2b5676f609d185fbda6acec16684b4fb6c3f758d6cf956a2ca8196b1ff1cece
   > Blocks: 0            Seconds: 0
   > contract address:    0x6D0014bDcDbf4aB01A651f27ebE42EFB053Bd868
   > block number:        1
   > block timestamp:     1793123185
   > account:             0x5641193d075314C86c3BF3742fe5e0511DA17535
   > balance:             99.999551317375
   > gas used:            132943 (0x2074f)
   > gas price:           3.375 gwei
   > value sent:          0 ETH
   > total cost:          0.000448682625 ETH

   > Saving artifacts
   -------------------------------------
   > Total cost:      0.000448682625 ETH

Summary
=======
> Total deployments:   1
> Final cost:          0.000448682625 ETH


truffle(develop)> █
```

Step 12: Test the deployed smart contract



```
truffle(develop)> let instance = await SimpleStorage.deployed()
undefined
truffle(develop)> await instance.set(100)
{
  tx: '0x208dff73068610a42077d6cabe4b2252522215ab98d35d148d9d6c5c99eb66c3',
  receipt: {
    transactionHash: '0x208dff73068610a42077d6cabe4b2252522215ab98d35d148d9d6c5c99eb66c3',
    transactionIndex: 0,
    blockNumber: 2,
    blockHash: '0x958ed0cdb225999d6efe24708dd220e9314afc7266f3b26d1c416b20be8c9538',
    from: '0x5641193d075314c86c3bf3742fe5e0511da17535',
    to: '0x6d0014bdcdbf4ab01a651f27ebe42efb053bd868',
    cumulativeGasUsed: 43718,
    gasUsed: 43718,
    contractAddress: null,
    logs: [],
    logsBloom: '0x0000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000',
    status: true,
    effectiveGasPrice: 3269951301,
    type: '0x2',
    rawLogs: []
  },
  logs: []
}
truffle(develop)> (await instance.get()).toString()
'100'
truffle(develop)> █
```

Result: The Truffle framework was studied, a local blockchain was established, and the `SimpleStorage` smart contract was successfully compiled, deployed, and tested by storing and retrieving the value 100.

Conclusion: Thus, the experiment was performed successfully using Truffle. The smart contract was deployed on a local blockchain, and its functionality was verified by storing and retrieving data.